PLC-Based Conveyor Automation: Verified Technical Report

Siemens TIA Portal V13 SP2 | S7-1200 CPU 1214C DC/DC/DC | WinCC Basic KTP700 Basic PN | PLCSIM / HMI RT Simulator

Evidence status: Documented from original 5-page OB1 LAD export (22 networks), previously provided HMI screenshots and recorded PLC traces. This is a simulated control application, not physical commissioning. "Verified" refers to fidelity to available evidence, not independent certification or a guarantee of fault-free operation.

1. System scope and operating sequence

Main OB1 controls the primary conveyor output %Q0.0, second conveyor %Q0.1, AUTO/MANUAL operation, START/STOP/JOG, virtual box movement, sensor-based counting, a ten-product target, sensor/motor fault latches and diagnostic flags. The HMI provides Home, Manual Control and Alarms screens. The PLC movement model is simulated; no real belt speed or physical sensor performance is established.

2. PLC logic — all 22 original networks

Network 1 — Motor latching

Normally closed STOP_PB %I0.1, HMI_STOP %M10.1, PRODUCT_COUNTER.QU, CONVEYOR_FAULT %M16.2 and MOTOR_FAULT %M16.5 form common permissives. AUTO_MODE %M17.2 permits START_PB %I0.2, HMI_START %M10.0 or CONVEYOR_MOTOR %Q0.0 seal-in. MANUAL_MODE %M17.3 permits MANUAL_RUN %M17.4 or JOG_CMD %M17.6, subject to normally closed MANUAL_STOP %M17.5. Output %Q0.0. This is a software command, not a safety-rated stop.

Network 2 — Product delay

PRODUCT_SENSOR %I0.3 energizes TON PRODUCT_TIMER DB1, PT = T#3s. Timer output Q is used in Network 3. This physical-input sensor is distinct from the virtual AUTO_SENSOR %M12.0.

Network 3 — Second conveyor

PRODUCT_TIMER.Q commands SECOND_CONVEYOR %Q0.1. It follows the delayed product sensor condition; the export does not establish additional mechanical sequencing or a second-conveyor interlock.

Network 4 — Product counter

CTU PRODUCT_COUNTER DB2: CU = AUTO_SENSOR %M12.0, R = RESET_COUNTER %I0.4 OR HMI_RESET %M10.2, PV = 10, CV = PRODUCT_COUNT %MW20. QU is used for target shutdown and alarm.

Network 5 — Movement timer

CONVEYOR_MOTOR %Q0.0 AND NOT MOVE_TIMER.Q enable TON MOVE_TIMER DB3, PT = T#1ms. Feedback causes cyclic timing behavior; actual pulse interval depends on scan timing and cannot be equated to exactly 1 ms per increment.

Network 6 — Movement edge

R_TRIG BOX_EDGE DB4 detects MOVE_TIMER.Q rising edge; Q drives MOVE_PULSE %M11.0.

Network 7 — Position increment

ADD Int executes when MOVE_PULSE %M11.0 is true: BOX_POSITION %MW30 = BOX_POSITION + 1.

Network 8 — Position wrap

When BOX_POSITION %MW30 >= 100, MOVE 0 to BOX_POSITION. HMI movement range uses this simulated position value.

Network 9 — Virtual sensor

AUTO_SENSOR %M12.0 is true when BOX_POSITION is between 42 and 80 inclusive AND SENSOR_FAIL_SIM %M16.3 is false. This is position-driven simulation, not the physical PRODUCT_SENSOR input.

Network 10 — Target alarm

PRODUCT_COUNTER.QU drives ALARM_TARGET_BIT %M15.0. Network 1 inhibits motor output when QU is true.

Network 11 — Sensor watch timeout

TON SENSOR_WATCH_TIMER DB5 uses SENSOR_MONITOR_ACTIVE %M17.1, PT = T#12s, Q = SENSOR_TIMEOUT %M16.0.

Network 12 — Sensor fault latch

SENSOR_TIMEOUT %M16.0 OR latched CONVEYOR_FAULT %M16.2, AND NOT FAULT_RESET %M16.1, drives CONVEYOR_FAULT %M16.2. Reset only clears if the driving condition no longer asserts.

Network 13 — Sensor alarm bit

CONVEYOR_FAULT %M16.2 drives ALARM_SENSOR_BIT %M15.1.

Network 14 — Motor fault latch

MOTOR_FAULT_SIM %M16.4 OR latched MOTOR_FAULT %M16.5, AND NOT FAULT_RESET %M16.1, drives MOTOR_FAULT %M16.5.

Network 15 — Motor alarm bit

MOTOR_FAULT %M16.5 drives ALARM_MOTOR_BIT %M16.6 (confirmed address).

Network 16 — Product transit window

PRODUCT_IN_TRANSIT %M17.0 = (BOX_POSITION >= 1) AND (BOX_POSITION < 42).

Network 17 — Sensor monitor enable

SENSOR_MONITOR_ACTIVE %M17.1 = CONVEYOR_MOTOR %Q0.0 AND (PRODUCT_IN_TRANSIT %M17.0 OR prior SENSOR_MONITOR_ACTIVE) AND NOT AUTO_SENSOR %M12.0. The self-holding branch maintains monitoring until the virtual sensor detects the product or the motor stops.

Network 18 — Manual mode complement

MANUAL_MODE %M17.3 = NOT AUTO_MODE %M17.2. The AUTO_MODE selection is driven externally via HMI events; the exported OB1 shows the complementary MANUAL_MODE assignment.

Network 19 — Combined fault flag

ANY_FAULT %M18.0 = CONVEYOR_FAULT %M16.2 OR MOTOR_FAULT %M16.5.

Network 20 — Start permission indication

START_PERMISSION %M18.1 = NOT MOTOR_FAULT AND NOT CONVEYOR_FAULT AND NOT ALARM_TARGET_BIT AND NOT STOP_PB. Network 1 does not directly use START_PERMISSION as its motor permissive.

Network 21 — Sensor missed indication

SENSOR_MISSED %M18.2 = (BOX_POSITION >= 42) AND NOT AUTO_SENSOR %M12.0. This may remain asserted beyond the expected detection window, depending on position and simulation state.

Network 22 — Short missed-sensor timer

TON SENSOR_MISS_TIMER DB6, PT = T#500ms, IN = SENSOR_MISSED %M18.2, Q = SENSOR_MISS_CONFIRMED %M18.3. No supplied OB1 rung connects this Q to SENSOR_TIMEOUT, CONVEYOR_FAULT, or motor stop.

3. HMI mapping and observed screens

HMI / feature	PLC signal / evidence
Home: START, STOP, RESET, FAULT RESET	HMI_START %M10.0, HMI_STOP %M10.1, HMI_RESET %M10.2, FAULT_RESET %M16.1
Home: product count and moving box	PRODUCT_COUNT %MW20; BOX_POSITION %MW30; animation observed in RT screenshots
Home: conveyor indicators	CONVEYOR_MOTOR %Q0.0; SECOND_CONVEYOR %Q0.1; virtual sensor %M12.0 shown in simulation
Manual: AUTO / MANUAL	AUTO_MODE %M17.2; MANUAL_MODE %M17.3; mode exclusivity supported by OB1 N18
Manual: RUN / STOP / JOG	MANUAL_RUN %M17.4; MANUAL_STOP %M17.5; JOG_CMD %M17.6
Manual: diagnostics	MOTOR_FAULT %M16.5, CONVEYOR_FAULT %M16.2, SENSOR_TIMEOUT %M16.0, START_PERMISSION %M18.1
Alarms	ALARM_TARGET_BIT %M15.0; ALARM_SENSOR_BIT %M15.1; ALARM_MOTOR_BIT %M16.6

The HMI displays an "EMERGENCY STOP OK" indicator in some screenshots. No corresponding wired E-stop or safety-rated function is established by OB1; do not claim E-stop validation. HMI discrete-alarm configuration is not fully proven by OB1 alone; runtime images demonstrate target and sensor messages, whereas motor-alarm popup mapping needs separate confirmation.

4. Test evidence and interpretation

Test	Observed evidence	Conclusion
Normal running	Trace: %Q0.0 ON, BOX_POSITION ramps/wraps, AUTO_SENSOR switches.	Simulation operation observed.
10-part target	Trace: count increments to 10, %Q0.0 drops, ALARM_TARGET_BIT rises; HMI shows production-target message.	Supported.
Sensor fault	HMI shows "Conveyor Sensor Timeout - Product Not Detected"; trace shows CONVEYOR_FAULT rise but SENSOR_TIMEOUT stays low.	Alarm observed; precise causal trace not confirmed.
Motor fault	Trace: MOTOR_FAULT_SIM rises, MOTOR_FAULT and ALARM_MOTOR_BIT assert, motor drops; reset clears and motor requires fresh start.	Supported by trace.
Manual controls	HMI screenshot shows mode selection and RUN/STOP/JOG.	Visual evidence; not a full timing trace.

5. Technical discrepancies / accuracy controls

Sensor trace mismatch: OB1 Network 12 derives CONVEYOR_FAULT from SENSOR_TIMEOUT and latch feedback. The provided sensor-fault trace shows CONVEYOR_FAULT asserting without SENSOR_TIMEOUT visibly asserting. The recording therefore does not independently demonstrate the expected 12 s timeout-to-latch transition. Possible causes require a new synchronized trace; none is asserted as fact.

500 ms timer: Networks 21–22 compute SENSOR_MISS_CONFIRMED, but it is not wired to the fault latch in this OB1 export. Therefore the current fault response must not be described as a 500 ms shutdown.

Second conveyor: The earlier report incorrectly said second-conveyor logic was not established. Networks 2–3 establish the 3 s TON and %Q0.1 output. No runtime trace of that output is provided.

Compilation: Screenshots show 0 errors and 10 warnings, including HMI button text warnings. Zero compile errors does not prove all alarm configurations and interlocks were functionally tested.

Safety: STOP_PB is a normal software contact in the shown ladder; no safety PLC, emergency-stop wiring, certified risk assessment, or machine-level validation is documented.

6. Recommended additional verification before external claims

Capture one synchronized trace with SENSOR_MONITOR_ACTIVE, SENSOR_WATCH_TIMER.ET/Q, SENSOR_TIMEOUT, CONVEYOR_FAULT, SENSOR_FAIL_SIM, MOTOR output, and FAULT_RESET. Verify alarm configuration for ALARM_MOTOR_BIT and test SECOND_CONVEYOR %Q0.1 against PRODUCT_SENSOR %I0.3 with its 3 s delay. These are pending tests, not completed results.

7. Portfolio-ready project description

Developed a Siemens S7-1200 PLC and KTP700 HMI conveyor automation simulation in TIA Portal V13 SP2. Implemented AUTO/MANUAL motor control, jog, virtual product movement, sensor-based counting to a 10-part target, a delayed second-conveyor output, sensor/motor fault latching, reset behavior and diagnostic screens. Validated core operating and fault-recovery behavior using PLCSIM, HMI runtime screenshots and PLC traces; documented remaining sensor-trace and alarm-mapping verification limits.

Evidence appendix: The following pages reproduce selected original HMI/trace evidence from the prior report, followed by the full unmodified 5-page OB1 ladder export (Networks 1–22). The original export is the authoritative reference for rung details.

10. Selected original visual evidence

The following six images are taken from the supplied evidence appendix. Captions have been checked against the visible screenshots; they are not recreated HMI screens.

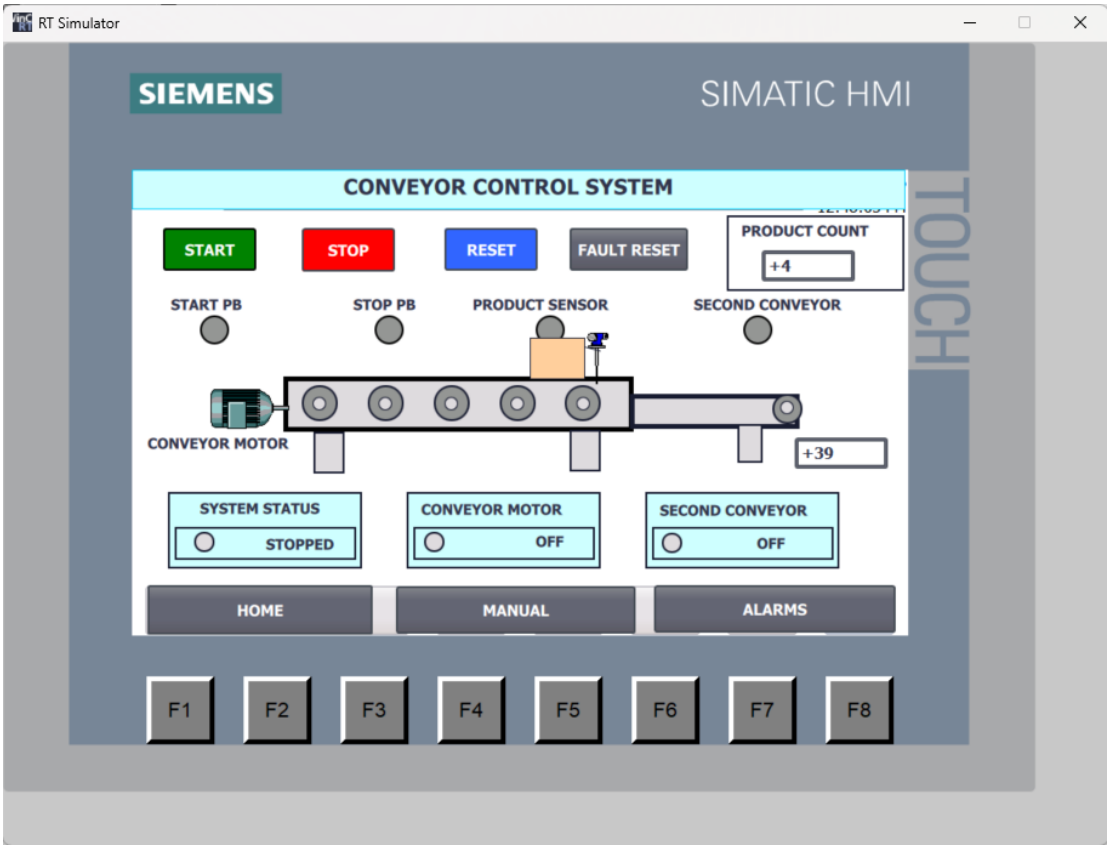


Figure 1. Home HMI — Conveyor overview with product count, START/STOP/RESET controls, animation, and equipment status.

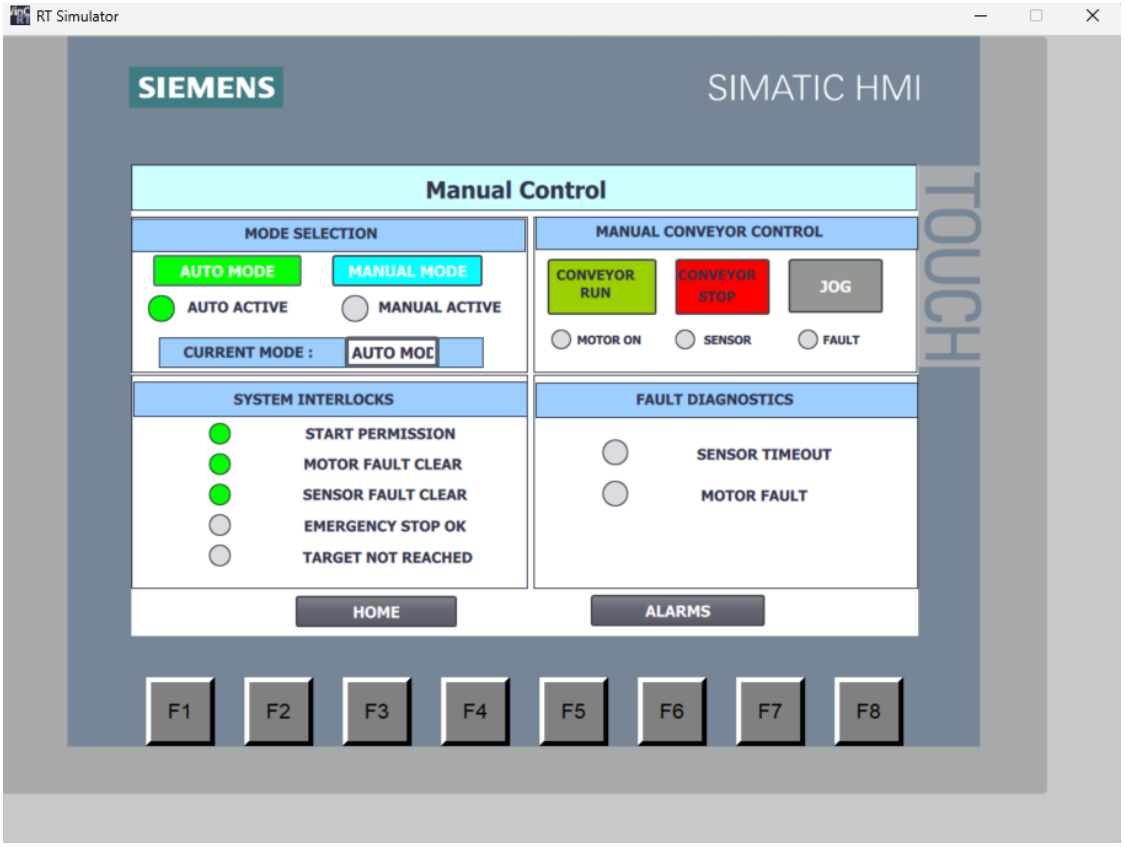


Figure 2. Manual Control HMI — AUTO/MANUAL selection, RUN/STOP/JOG controls, interlock indicators, and fault diagnostics.

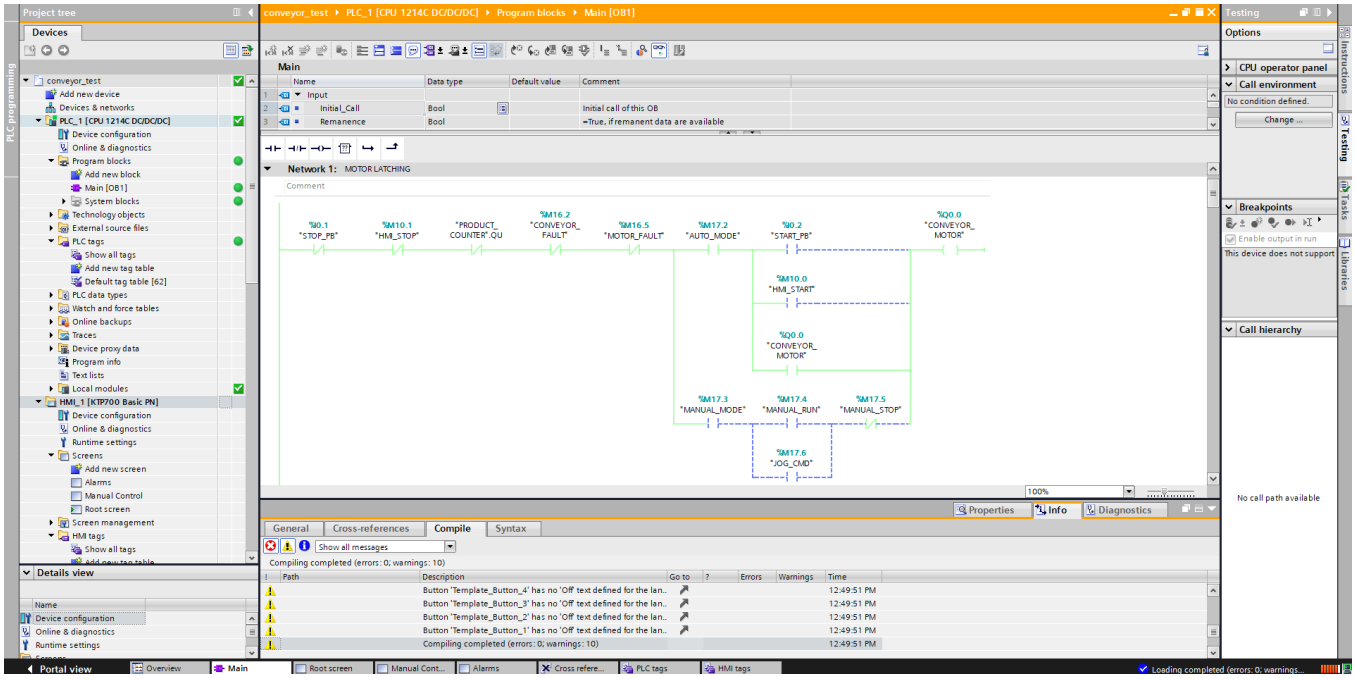


Figure 3. PLC motor-control ladder — Main OB1 motor latching network with mode branches, target-complete and fault interlocks.

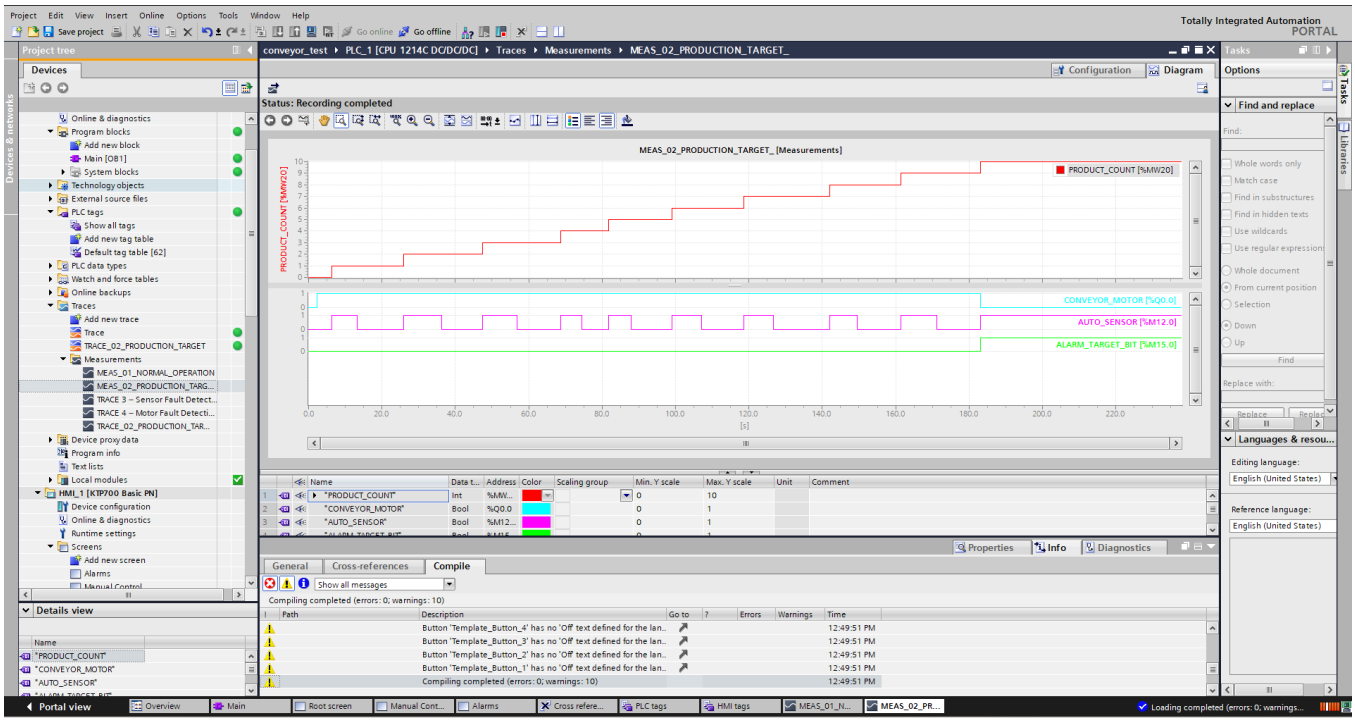


Figure 4. Production-target trace — Count increases to 10; motor output drops and ALARM_TARGET_BIT activates at target.

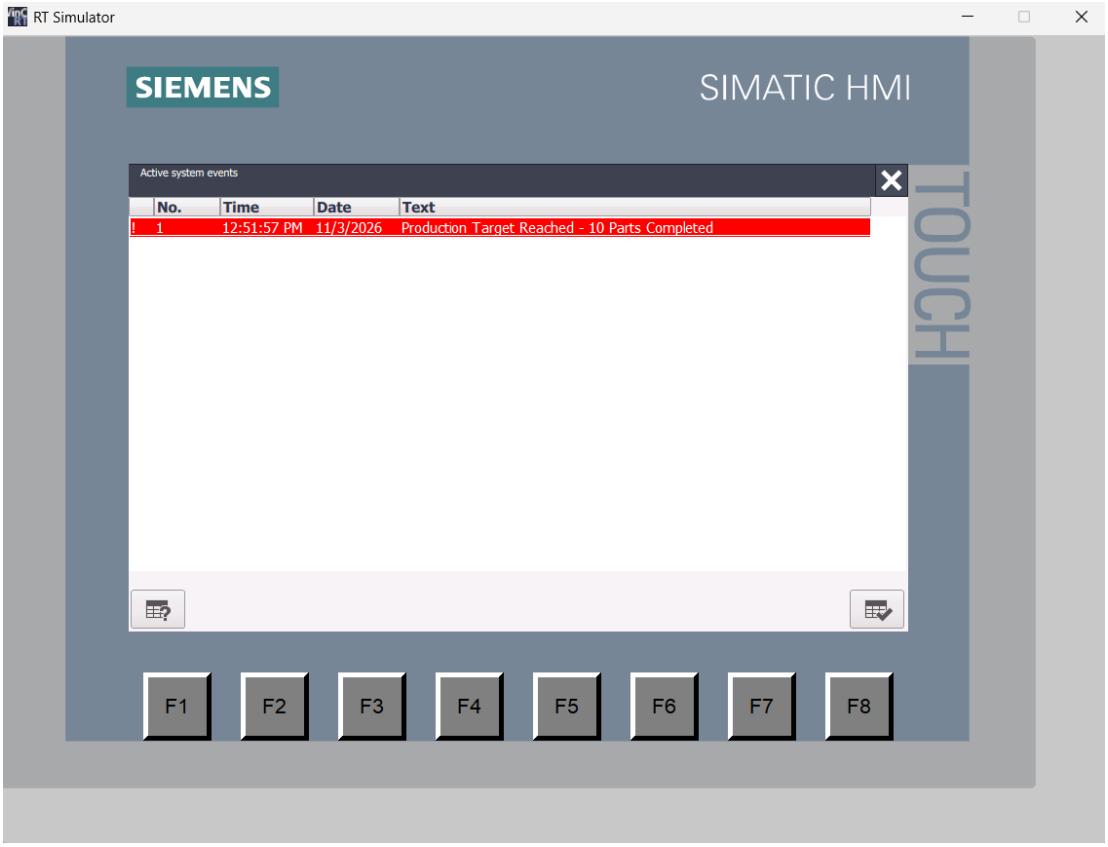


Figure 5. Production target alarm — Actual visible message: “Production Target Reached - 10 Parts Completed.” This is not a sensor-timeout screenshot.

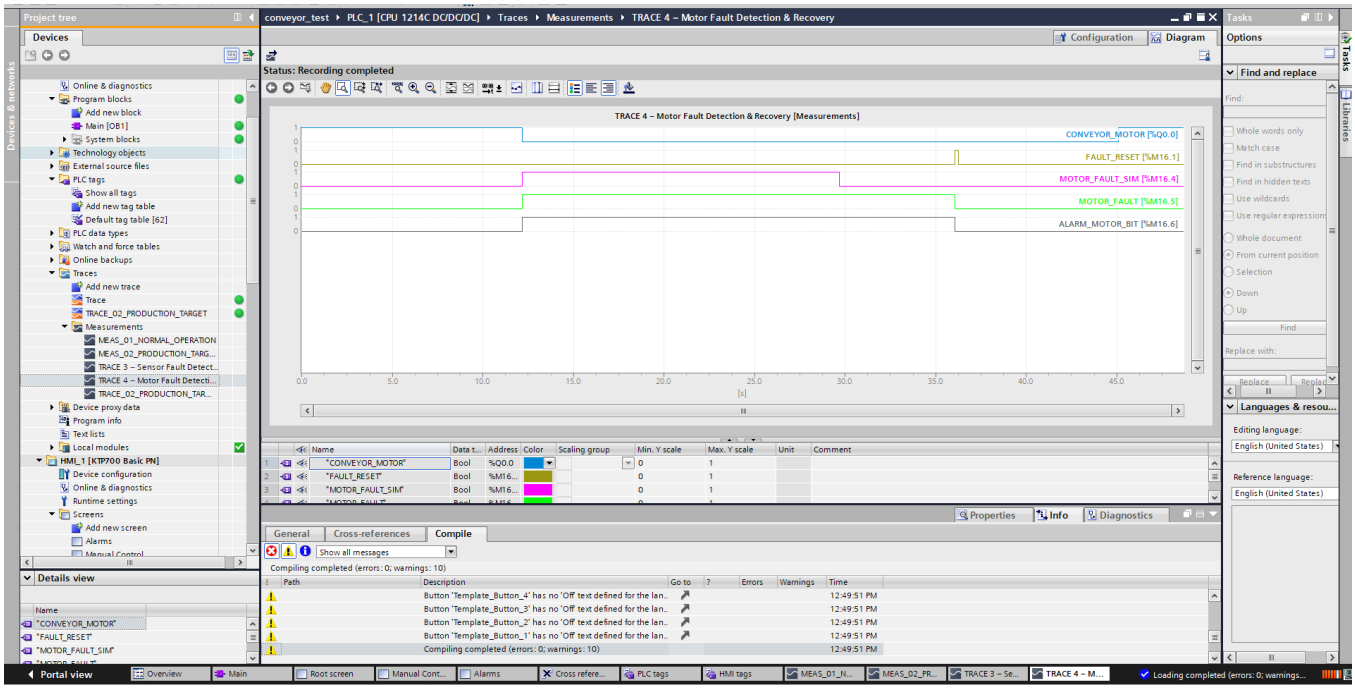


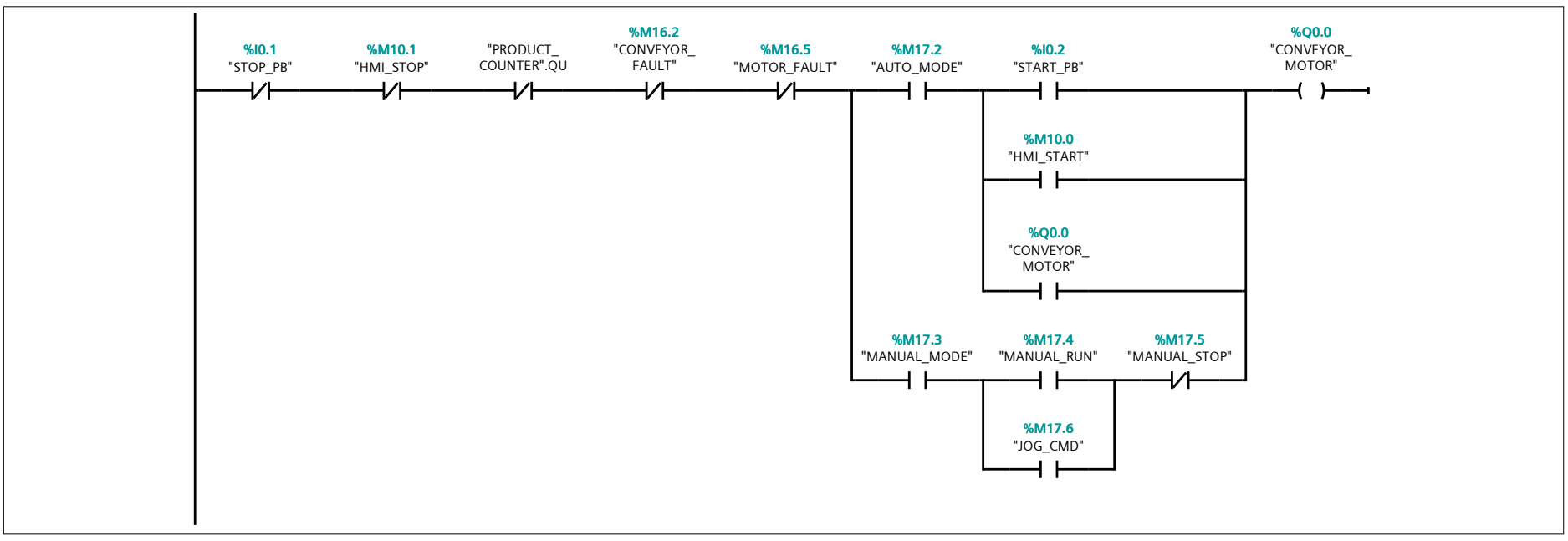
Figure 6. Motor-fault recovery trace — MOTOR_FAULT_SIM and MOTOR_FAULT rise; motor drops; reset clears fault and alarm without automatic restart.

Main [OB1]

Main Properties							
General							
Name	Main	**Number**	1	**Type**	OB	**Language**	LAD
Numbering	automatic						
Information							
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

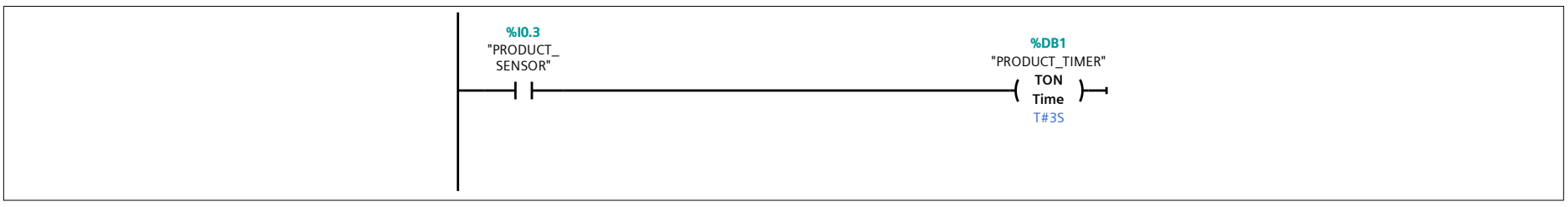
Name	Data type	Default value	Comment
Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

Network 1: MOTOR LATCHING



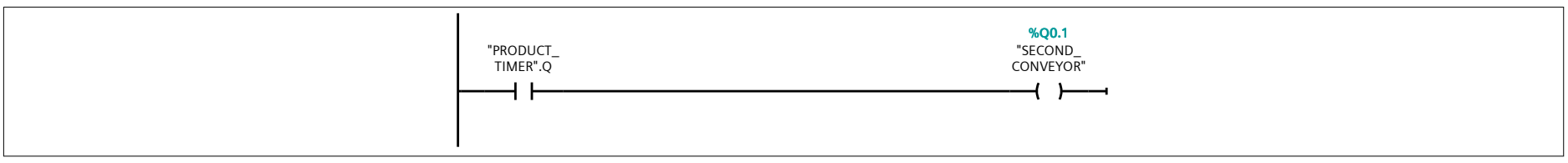
Symbol	Address	Type	Comment
"AUTO_MODE"	%M17.2	Bool	
"CONVEYOR_FAULT"	%M16.2	Bool	
"CONVEYOR_MOTOR"	%Q0.0	Bool	CONVEYOR MOTOR
"HMI_START"	%M10.0	Bool	HMI_START
"HMI_STOP"	%M10.1	Bool	
"JOG_CMD"	%M17.6	Bool	
"MANUAL_MODE"	%M17.3	Bool	
"MANUAL_RUN"	%M17.4	Bool	
"MANUAL_STOP"	%M17.5	Bool	
"MOTOR_FAULT"	%M16.5	Bool	
"PRODUCT_COUNTER".QU		Bool	
"START_PB"	%I0.2	Bool	CONVEYOR START PB
"STOP_PB"	%I0.1	Bool	CONVEYOR STOP PB

Network 2: PRODUCTION AND DELAY



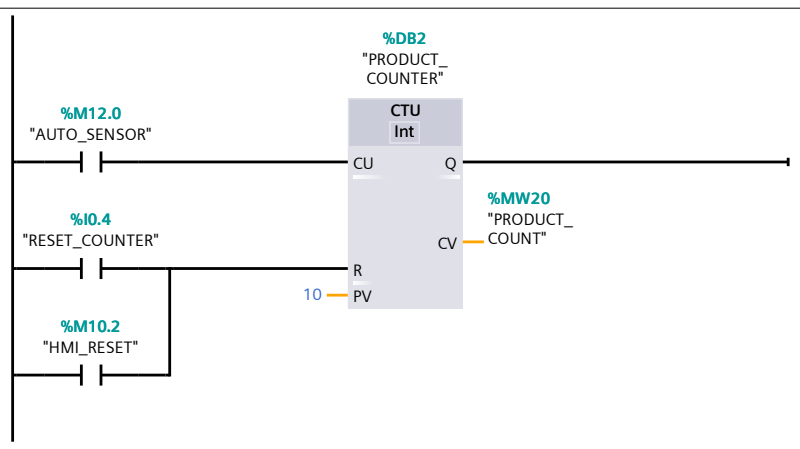
Symbol	Address	Type	Comment
"PRODUCT_SENSOR"	%I0.3	Bool	PRODUCT SENSOR

Network 3: SECOND CONVEYOR



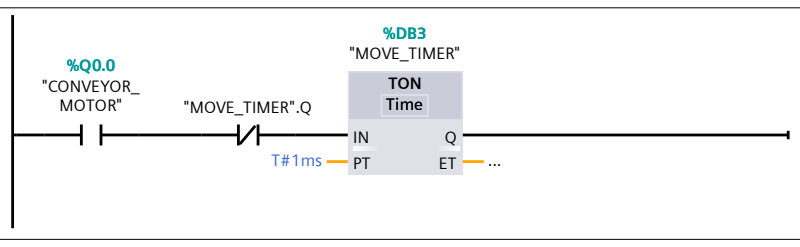
Symbol	Address	Type	Comment
"PRODUCT_TIMER".Q		Bool	
"SECOND_CONVEYOR"	%Q0.1	Bool	SECOND CONVEYOR

Network 4:



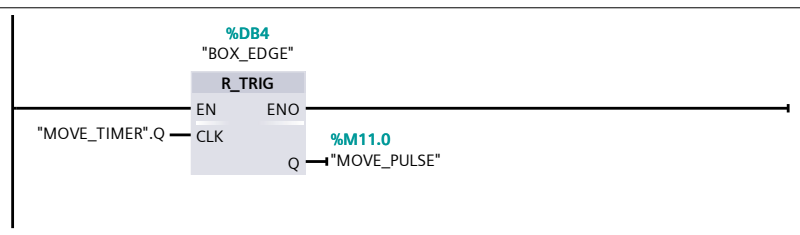
Symbol	Address	Type	Comment
"AUTO_SENSOR"	%M12.0	Bool	
"HMI_RESET"	%M10.2	Bool	
"PRODUCT_COUNT"	%MW20	Int	PRODUCT COUNTER
"RESET_COUNTER"	%I0.4	Bool	RESET COUNTER

Network 5:



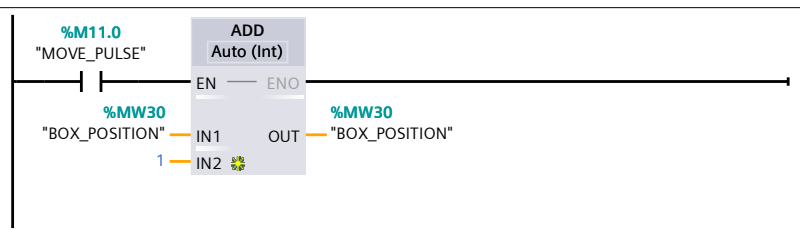
Symbol	Address	Type	Comment
"CONVEYOR_MOTOR"	%Q0.0	Bool	CONVEYOR MOTOR
"MOVE_TIMER".Q		Bool	

Network 6:



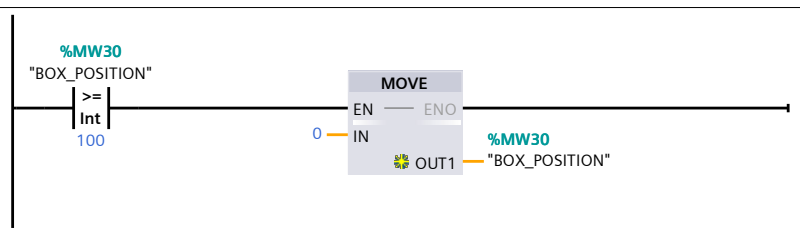
Symbol	Address	Type	Comment
"MOVE_PULSE"	%M11.0	Bool	
"MOVE_TIMER".Q		Bool	

Network 7:



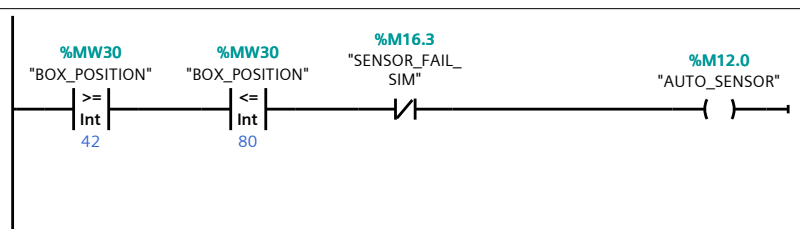
Symbol	Address	Type	Comment
"BOX_POSITION"	%MW30	Int	
"MOVE_PULSE"	%M11.0	Bool	

Network 8:



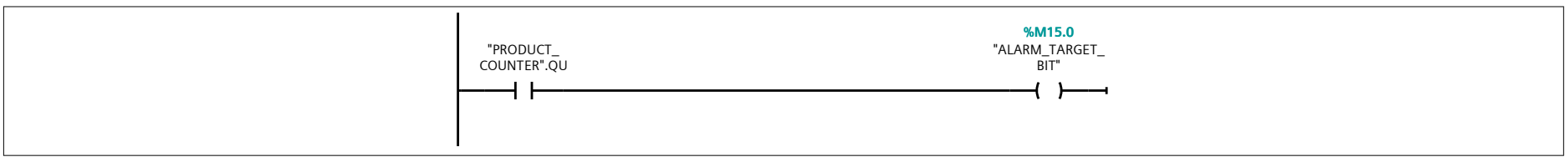
Symbol	Address	Type	Comment
"BOX_POSITION"	%MW30	Int	

Network 9:



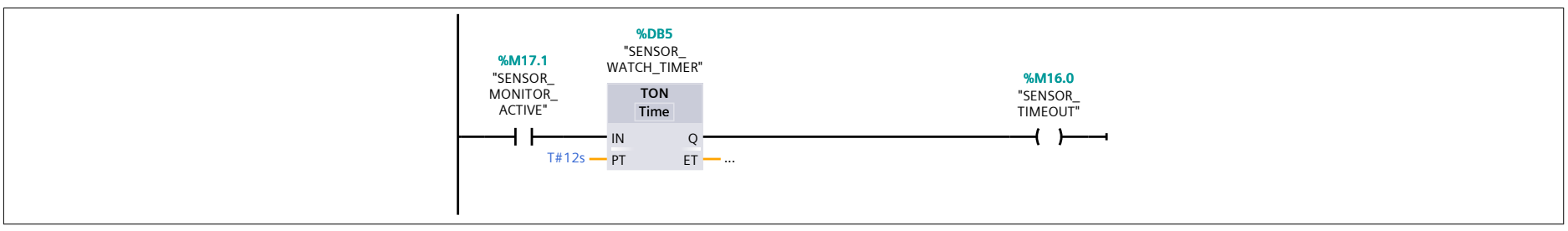
Symbol	Address	Type	Comment
"AUTO_SENSOR"	%M12.0	Bool	
"BOX_POSITION"	%MW30	Int	
"SENSOR_FAIL_SIM"	%M16.3	Bool	

Network 10:



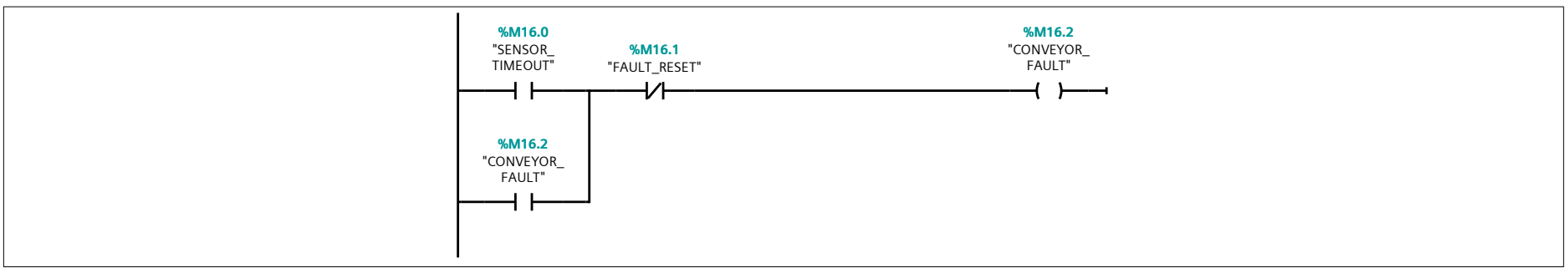
Symbol	Address	Type	Comment
"ALARM_TARGET_BIT"	%M15.0	Bool	
"PRODUCT_COUNTER".QU		Bool	

Network 11:



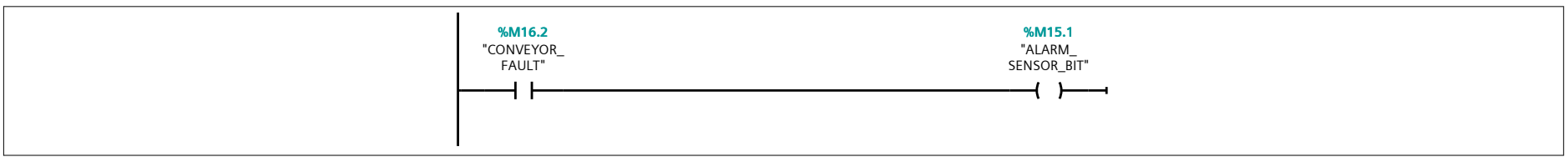
Symbol	Address	Type	Comment
"SENSOR_MONITOR_ACTIVE"	%M17.1	Bool	
"SENSOR_TIMEOUT"	%M16.0	Bool	

Network 12:



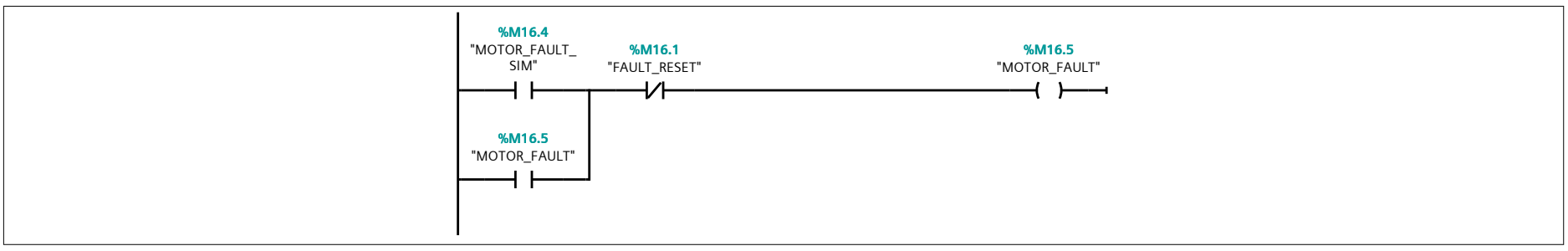
Symbol	Address	Type	Comment
"CONVEYOR_FAULT"	%M16.2	Bool	
"FAULT_RESET"	%M16.1	Bool	
"SENSOR_TIMEOUT"	%M16.0	Bool	

Network 13:



Symbol	Address	Type	Comment
"ALARM_SENSOR_BIT"	%M15.1	Bool	
"CONVEYOR_FAULT"	%M16.2	Bool	

Network 14:



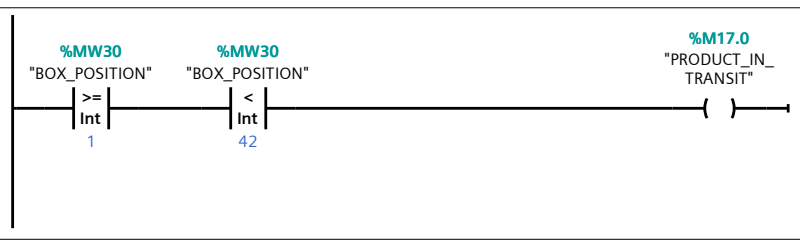
Symbol	Address	Type	Comment
"FAULT_RESET"	%M16.1	Bool	
"MOTOR_FAULT"	%M16.5	Bool	
"MOTOR_FAULT_SIM"	%M16.4	Bool	

Network 15:



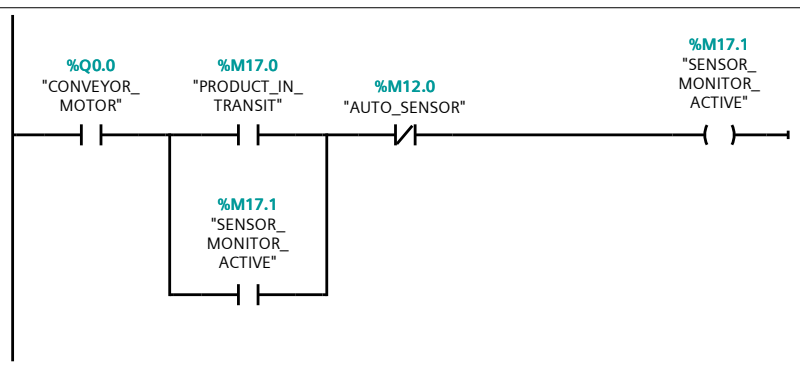
Symbol	Address	Type	Comment
"ALARM_MOTOR_BIT"	%M16.6	Bool	
"MOTOR_FAULT"	%M16.5	Bool	

Network 16:



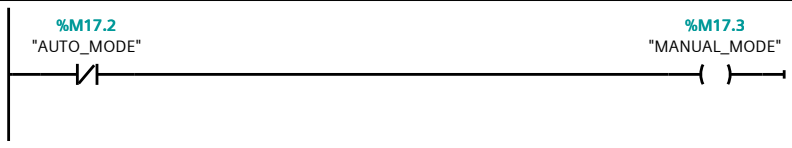
Symbol	Address	Type	Comment
"BOX_POSITION"	%MW30	Int	
"PRODUCT_IN_TRANSIT"	%M17.0	Bool	

Network 17:



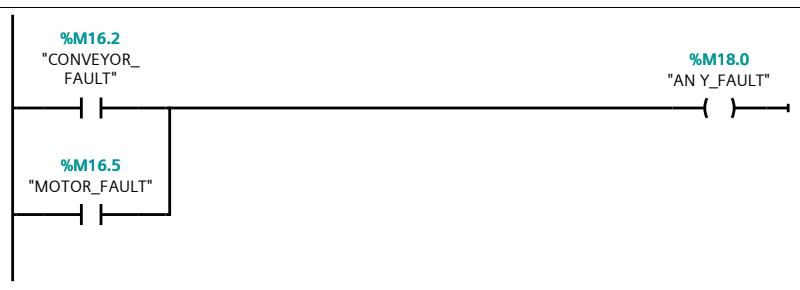
Symbol	Address	Type	Comment
"AUTO_SENSOR"	%M12.0	Bool	
"CONVEYOR_MOTOR"	%Q0.0	Bool	CONVEYOR MOTOR
"PRODUCT_IN_TRANSIT"	%M17.0	Bool	
"SENSOR_MONITOR_ACTIVE"	%M17.1	Bool	

Network 18:



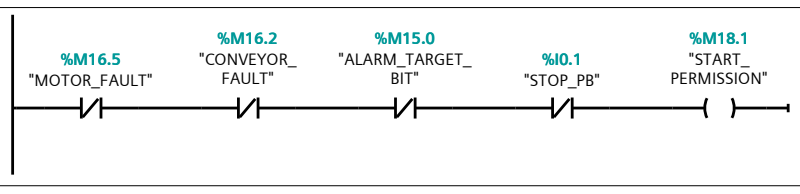
Symbol	Address	Type	Comment
"AUTO_MODE"	%M17.2	Bool	
"MANUAL_MODE"	%M17.3	Bool	

Network 19:



Symbol	Address	Type	Comment
"AN Y_FAULT"	%M18.0	Bool	
"CONVEYOR_FAULT"	%M16.2	Bool	
"MOTOR_FAULT"	%M16.5	Bool	

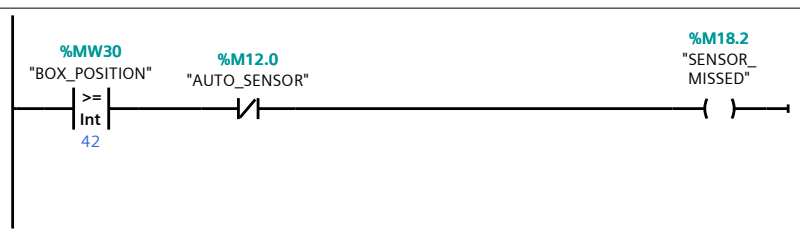
Network 20:



Symbol	Address	Type	Comment
"ALARM_TARGET_BIT"	%M15.0	Bool	
"CONVEYOR_FAULT"	%M16.2	Bool	
"MOTOR_FAULT"	%M16.5	Bool	

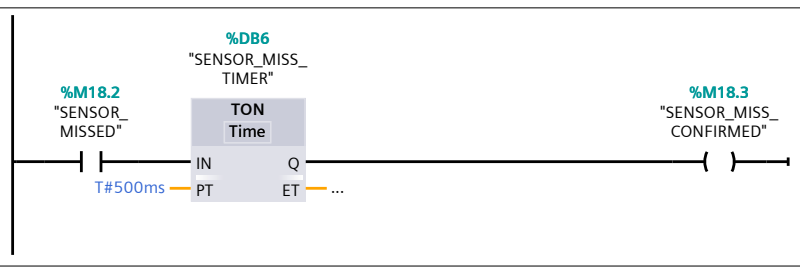
Symbol	Address	Type	Comment
"START_PERMISSION"	%M18.1	Bool	
"STOP_PB"	%I0.1	Bool	CONVEYOR STOP PB

Network 21:



Symbol	Address	Type	Comment
"AUTO_SENSOR"	%M12.0	Bool	
"BOX_POSITION"	%MW30	Int	
"SENSOR_MISSED"	%M18.2	Bool	

Network 22:



Symbol	Address	Type	Comment
"SENSOR_MISS_CONFIRMED"	%M18.3	Bool	
"SENSOR_MISSED"	%M18.2	Bool	